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Project Title:	Complex Plasma under Microgravity, Utilizing the International Space Station Experiment “PK-4” and Beyond		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
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Space Biology Element:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
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No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2023 per U. Israelsson/JPL (Ed., 1/6/22) NOTE: End date changed to 12/31/2021 per U. Israelsson/JPL (Ed., 4/13/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thomas, Edward Ph.D. (Auburn University) Williams, Jeremiah Ph.D. (Wittenberg University)		
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Task Description:

Complex plasmas are four-component systems that contain electrons, ions, neutral atoms, and charged microparticles (i.e., the dust particles). Depending upon the experimental conditions, the microparticles in these systems can show solid-like, liquid-like, or gas-like properties. However, ground-based experiments are subject to gravitational forces that often dominate the behavior of the systems under investigation leading to sedimentation and anisotropic effects. Therefore, in order to study large-scale complex plasmas and the influences of the weaker forces that are masked by the gravitational force, experiments have to be performed under “microgravity” conditions. Over the past decade, several complex plasma experiments have been performed using the International Space Station (ISS). The latest of these research instruments, “Plasma Kristall 4” (PK-4), is the first microgravity laboratory that is optimized to study fluid phenomena and fundamental effects such as particle charging, screening, and the influence of a directed ion flow. Additionally, since PK-4 is a multi-user facility, we have a unique opportunity to use this instrument to study ion/particle and particle/particle interactions that we otherwise would not be able to address in ground-based experiments.
The intellectual merit of this research project is centered on three major research topics.

Topic 1: Charging and interaction of microparticles in the plasma. The most important measurement in a complex plasma is the measurement of the charge on the particles. However, direct experimental measurement of the charge is difficult, because it usually relies on additional assumptions about experimental parameters such as the plasma electric field and on the validity of an assumed particle interaction type. The microgravity environment of PK-4, where the dust is suspended in the bulk plasma, allows experiments that can potentially lead to a higher accuracy in the particle charge determination.

Topic 2: Ion-dust interactions and the thermal properties of dusty plasmas. A key observation in complex plasmas is the unusually high dust kinetic temperature. It has been postulated that ion-dust interactions are largely responsible for heating the dust particles. The microgravity environment of the PK-4 laboratory, in combination with the tools to actively manipulate the ions in the plasma, provides a unique platform for studying the transfer of energy between ions and the microparticles in the plasma.

Topic 3: Particle transport and heating in waves. The low charge to mass ratio of the dust leads to characteristic frequencies for the dust component that are on the order of a few Hz. This, coupled with the finite size of the dust grains, allows researchers a unique opportunity to directly image plasma physics phenomena and particle transport at the kinetic level. In the microgravity environment of PK-4, it will be possible to measure the transport properties of particles during their initial injection into the bulk plasma, follow the particles as they form collective structures (e.g., waves), and study how ion flow modifies the thermal properties of these waves.

This proposed research project has several opportunities to have a broader impact: (1) through on- and off-campus outreach activities and the training of undergraduate and graduate students, this project will cultivate the next generation of plasma scientists; (2) through the development of new ground-based laboratory instruments (e.g., plasma sources, diagnostics) that will complement the proposed research and contribute to the development of next-generation dusty plasma microgravity experiments; and (3) through the development of new, advanced software tools for image analysis and particle tracking that will support the proposed studies and be offered as community-wide research tools.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research performed under this project falls in the main category of fundamental research. Often, findings obtained by performing fundamental research show their true impact for the society only after many years. However, the study of dusty/complex plasma can benefit the overall research infrastructure in many ways already in the timely vicinity of any discoveries made, mainly as a result of the strong interdisciplinary nature of this research. Studying complex plasma topics such as wave phenomena, heating, the spatial separation of phases, their interaction or the transition between phases such as from fluids to crystals all being analyzed down to the scale of individual particles can lead to fundamental findings that might be immediately applied to many other research fields such as solid state physics, fluid physics, plasma physics, astrophysics. As the specific research performed within the scope of this project makes use of the microgravity (free-fall) environment supported by the national laboratory in space, the International Space Station (ISS), further utilizing an experiment that is built by the European Space Agency (ESA), highly international cooperation activities are a direct result. It is likely that the effort to work on the same fundamental questions in a cooperative acting international team will ultimately benefit each involved nation also beyond just the research. Some aspects of the specific research topics within the scope of this project involve the development of universal software, hardware, or analysis tools that can also be used under different research situations, and some of the more specific technical tools that are developed, such as a plasma impedance probe, might even help to overall improve technology that can be used for industrial purposes. Finally, as part of the general research mission, we train the next generation of engineers and scientists who can in the course of their future career produce a large benefit for the overall economic situation of the nation and beyond.

The fourth year of our project started on April 16th 2020. The team, a cooperation between Auburn University and Wittenberg University, continued working in about the same constellation as during the last project year. In the fall of 2020 we added one undergraduate student to our team at Auburn University, funded via the NSF (National Science Foundation) part of the project. During the reporting period we continued our fruitful cooperation with the three external major institutions: 1) the leading PK-4 core group of the International Space Station (ISS) experiment “Plasma Kristall 4” (PK-4), localized at the German Aerospace Center (DLR) in Wesslingen, Germany; This group operate the ISS based experiment PK-4 as well as a ground based reference module and also perform parabolic flights with a potential future successor complex plasma experiment, COMPACT. 2) Baylor University in Texas. Here a PK-4 equivalent ground setup is accessible to our group within the cooperation. In addition, we usually proposed PK-4 experiments in a team with Baylor University to make use of synergies. 3) Giessen University, Germany. The group at Giessen invited us to participate in a parabolic flight campaign using a PK-4 equivalent experiment as well as to use some of their prior obtained data.
For 2020 it was planned that the group would take part on two parabolic flight campaigns, one in May performed by Giessen University utilizing a PK-4 ground based experiment, and one in June as part of a campaign organized by the German Aerospace Center utilizing a breadboard setup of the next generation space experiments for complex plasmas, COMPACT. Also, substantial work directly at the DLR and the Giessen group was organized in detail to perform experiment at their laboratories as part of a sabbatical semester of the Principal Investigator (PI). These and other planned activities that involved traveling had to be postponed and often even finally canceled as a result of the world

Task Progress:

wide COVID-19 pandemic that made traveling internationally and even nationally practically impossible. As such, beyond the major planned but ultimately canceled activities of the PI at DLR and Giessen, the participation of most team members at several (international) meetings was also postponed or canceled. For example, the major dusty plasma conference International Conference on the Physics of Dusty Plasmas (ICPDP) (3 annual conference) planned to be held in Moscow was postponed to 2022. Also, as a side effect of the pandemic, the delivery of the experimental data from our experiments aboard the ISS from campaign 9 has been postponed. Instead of a delivery in the summer 2020 as expected the data delivery is now projected to arrive at the end of this reporting period, April 2021. Overall, the pandemic influence has manifested itself on our work capabilities, by travel restrictions, restrictions on operation of the lab with one researcher at a time and delivery delays for equipment and data.

To adapt to the imposed restriction of the COVID pandemic, a modification to our original planned activities has been implemented early on. Our experimental work locally at Auburn University has been prioritized to plasma diagnostic measurements by Brandon Doyle (PhD student) that is expected to allow to measure the average charging state of a dusty plasma when fully developed. Other topics have been studied by developing theoretical models, and analyzing data from former experiments. Experimental data that have been studied had been obtained using the PK-4 and their equivalent ground and parabolic flight setups. Also numerical simulations were performed and our analysis, computation tools were improved. The travel restrictions were overcome using online messaging tools (Zoom and others) to setup ad-hoc group meetings and participating in major virtual held conference in the fall. We participated in conferences such as the American Physical Society (APS) meeting for the division of plasma physics, the American Society for Gravitational & Space Research (ASGSR) meeting as well as the ISS Research & Development Conference (ISSRDC). Overall we contributed to national and international meetings with 11 talks/poster in the reporting period. A special opportunity evolved, partly because of the pandemic, and the globally introduced online virtual meeting capabilities. As part of our extension activities towards the realization and definition of a follow up ISS based complex plasma space experiment, the PI was invited to join the science definition team for COMPACT, the COMplex PlAsma faCiliTy. This teams is a result of a joint effort of multiple (space) organizations including DLR, NASA, ESA, NSF, and ROSCOSMOS. As part of the activities, the PI organized and hosted at Auburn (virtually) in cooperation with the German Aerospace Center (DLR) an international science definition workshop for COMPACT with more than 80 participants (count does not include agency representatives) discussing potential science objectives for this potential future space experiment on the bases of 24 international, individual contributions (talks). Beside the mostly negative influences of the pandemic, the international communication was actually enhanced via the enhanced available tools for online communication, which we will continue to use in the future.

In the following we give a short, topic specific summary of the activities/progress as archived in year four of this project towards its mission to gain new knowledge in the physics of complex (dusty) plasmas and to create a vision beyond the ongoing experiments towards future space experiments on complex plasmas in space (presumably aboard the ISS). The research that has been conducted is categorized in three major sub topics:

Topic T1: Charging and interaction of micro-particles in a plasma, Topic T2: Ion-dust interaction and the thermal properties of dusty plasmas, Topic T3: Particle transport and heating in waves, Topic TX: Future space experiments, administrative tasks, and analysis tool developments.

Despite the pandemic related restrictions, we were able to archive the following progress towards our project goals: Topics T1: As part of an approach to determine the average charge of each particle in a dusty plasma based on the overall charge neutrality assumption, we studied within a PhD framework a radio frequency (rf) attenuation/reflection measurement technique to determine the electron plasma density, which will be combined in the final step with an ion density determination from the ion saturation current measurement using the same probe. Our progress included the development of a plasma/probe equivalent circuit model based on an analytical plasma fluid model. We performed multiple experiment series to measure the frequency variation in the plasma impedance for a variety of plasma conditions. From the impedance characteristics we started to derive the electron plasma density using our before developed analytical plasma circuit model. In a second approach to determine the particle charge in a plasma we have studied the density variation of a streaming dust cloud in a dusty plasma under the influence of a magnetic field. In the reporting year, our analytical approach based on a dust fluid model has been improved. It now connects the dust number density gradient to the average dust particle charge with an analytical expression. The theory is under test using molecular dynamic simulations that also have been improved in the recent year. The simulation needs further improvement to allow to validate the theory for a broader parameter range though. We expect that the theoretical model that is based on a fluid approach will break down for systems of strong coupling that are close to the liquid solid phase transition. A separate, limiting theory will potentially be derived for this limit in the coming year. Due to the COVID related restriction and thus the cancellation of the PI’s work at DLR and Giessen University, which were supposed to include the participation in two parabolic flight campaigns, the third approach to measure dust charges in a dusty plasma based on dust-dust collision analysis did not make a significant process in 2020.

Topic T2/T3: Topics T2 and T3 are performed at a reduced scale compared to the initial proposal due to lower funding and thus under a reduced manpower. As such we report on the effort on these two topics combined. The reporting year started just after we participated in our second experiment campaign utilizing the ISS PK-4 experiment, controlled from CADMOS in Toulouse, France. The data of this campaign was expected to arrive in the mid of 2020. Due to COVID related travel restriction the data has not been delivered to Auburn yet but is expected by April 2021. We worked mostly on the analysis of data obtained during our first PK-4 campaign (PK-4 campaign 7) as well as data that we obtained from our participation in a parabolic flight campaigns in cooperation with Giessen University. We studied especially the details of the energy distribution in the dusty plasma shortly after the onset of polarity switching. Initially a dust cloud introduced in PK-4 is moved with the ion flow along the discharge tube under the influence of a static electric field. To stop the motion of the dust cloud in the field of view of a double camera system, the discharge mode is switch to a polarity switch mode where the time averaged background electric field vanishes along the tube axis. Under the alternating ion flow direction the dust-dust interaction changes and thus the force balance in the system is disrupted instantly. On the way of finding a new force balance particles are accelerated and rearranged and the initial directed motion is randomized. We studied the details of the spatial, temporal and the isotropic evolution of the kinetic temperature during this process.

Topic TX: Beside the already mentioned activities to define the science of a future followup space experiment to PK-4, the experiment facility “COMPACT,” we started the analysis of an experiment we performed with the ground based breadboard model at DLR. In this experiment a large dust cloud was generated and a phase transition from a liquid state to a solid state was triggered by compressing the system under the motion of the confining electrodes and thus the reduction of the plasma volume. We used this experiment for an undergraduate student project to get him familiar with

and expand our used dusty plasma analysis techniques. The undergraduate student Mason Sake has been hired in the fall of 2020 to perform this tasks. During the few months of work on this project he has already succeeded to setup an analysis tool that allows to reasonably discriminate local crystalline structures with respect to their “crystallization strength” and the local crystallization orientation in space from the positional data of the particles in the system. This project will be continued beyond the project scope.

Overall: We presented our progress, results, and ideas in the fourth year at multiple virtual meetings. Results were presented for example during the 36th Annual ASGSR meeting, the Division of Plasma Physics annual meeting of the American Physical Society, as well as the International Space Station Development and Research Conference. Due to the pandemic most outreach activities were canceled. However, worth to mention, the PI helped to found the “Astronomical Society at Auburn University” in spring 2020 and has held the position as an advisor for the society since then. Overall the scientific process is somewhat less strong than we hoped for, but more significant than the pandemic might have led to. Our flexible research approach still has let us proceed with a significant process.

An outlook for PK-4 and beyond: To compensate for the deviation from the original research plan as a result of the pandemic in 2020, we requested a no-cost extension of this project. The extension will allow us to bring multiple aspects of our work plan to its final goals. As PK-4 will be also operational for a couple of more years due to the extended support by the space agencies, this further opens up the possibility to gain additional experimental data from the ISS facility by participation in additional experiment campaigns. On the near time scale, we also plan to explore additional research questions of complex/dusty plasmas by utilizing PK-4 within a new project frame, since PK-4 momentarily is the only, world-wide available complex/dusty plasma experiment that allows to study these systems under reduced influence of the gravitational acceleration. Multiple activities might be proposed and performed as long as PK-4 will be operational aboard the ISS.

In parallel to utilizing the PK-4 experiment we are incorporated into an international team of scientists, distributed across the globe, to define and realize a next generation complex plasma space experiment that is based on the formerly development of “Plasmalab” alias “Ekoplasma” now reborn under the working title “COMPACT,” a synonym for “COMplex PlAsma faCiliTy.” We will continue our activities as a major international contributor to realize this experiment and aim to open it up for utilization to the broader scientific community reaching beyond the core complex/dusty plasma community. Since the PI has been for many years the lead scientist/developer and manager of this upcoming, top-notch facility during its early stages, we expect that our team, depending on agency support, might in the future co-lead the development of and studies with COMPACT. The latter depends on the future support for the experiment realization and operation aboard the ISS as part of an international partnership between the world-wide leading space agencies. We are looking forward to a promising, due to COVID though still challenging, eventful extension of the project in the coming year.

Bibliography Type:	Description: (Last Updated: 09/30/2025)
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